KAZAKH-BRITISH TECHNICAL UNIVERSITY

KBTU

JSC «Kazakh-British Technical University»
School of IT and Engineering

APPROVED BY
Dean of SITE
Azamat Imanbayev
« ___ » ____________ **2025**

SYLLABUS

Discipline: **Fundamentals of Business for Information Systems**
Number of credits: 3
Course code – INFT3106
Term: Fall 2025
Instructor full name: Dr. Avinash Bigadevanahalli Mruthyunjaya

Personal Information about the Instructor	Time and place of classes		Contact information	
	Lessons	Office Hours	Tel.	e-mail
Dr. Avinash BM, PhD. Associate Professor, School of Information Technology and Engineering	**Monday** - Batch 1: 09:00 AM to 12:00 PM, 726 **Tuesday** - Batch 2: 04:00 PM to 07:00 PM, 461 **Wednesday** - Batch 3: 10:00 AM to 01:00 PM, 222 **Wednesday** - Batch 4: 02:00 PM to 05:00 PM, 222 **Thursday** - Batch 5: 11:00 AM to 02:00 PM, 446 **Thursday** - Batch 6: 03:00 PM to 06:00 PM, 414	Room 274 **Mon** 12 to 01 PM Offline and 2 to 6 Online **Tue** 10 to 02 Online **Wed** 01 to 02 Offline **Thu** 02 to 03 Offline **Fri** 10 to 01 Online	+77056323079 WhatsApp MS Teams	a.bigadevanahalli @kbtu.kz

COURSE DURATION: 3 credits, 15 weeks, 45 class hours

GENERAL COURSE AIMS: The course aims to

1. To develop an understanding of how information systems support and transform global business operations.
2. To equip students with knowledge of enterprise applications and digital platforms.
3. To foster critical thinking in aligning information systems with organizational strategy and decision-making.
4. To prepare students for real-world challenges in managing and building information systems.

COURSE DESCRIPTION:

This course introduces students to the fundamentals of business in the context of information systems, emphasizing their role in driving global competitiveness and innovation. It examines how enterprise applications, such as ERP, SCM, and CRM, enhance operational excellence and customer intimacy. Students will gain insights into e-commerce, digital markets, and the strategic alignment of IS with organizational goals. The course further examines global collaboration, decision-making support, and system development practices. By integrating theory with real-world cases, students will understand how to design, manage, and leverage information systems to address complex business challenges in a digital economy.

COURSE OBJECTIVES: The objectives of the course are to

1. To provide foundational knowledge of information systems in business by examining their role in global operations, competitiveness, and digital transformation.
2. To develop an understanding of enterprise applications such as ERP, SCM, CRM, and e-commerce platforms that enhance operational excellence and customer intimacy.
3. To cultivate analytical and strategic thinking in aligning information systems with organizational goals, decision-making processes, and collaborative practices.
4. To prepare students for real-world application of designing, building, and managing information systems in global and digital business environments.

COURSE OUTCOMES: By the end of this course, students will be able to:

1. Explain the role of information systems in global business and their impact on operational efficiency, competitiveness, and innovation.
2. Analyze and evaluate enterprise applications (ERP, SCM, CRM, and e-commerce platforms) for improving business processes and customer engagement.
3. Apply critical thinking to align information systems with organizational strategy, fostering collaboration and informed decision-making in diverse business contexts.
4. Design and propose solutions for managing, building, and leveraging information systems to address real-world challenges in the digital economy.

COURSE POST REQUISITES: After completing this course, students will be able to

1. Pursue advanced courses in Enterprise Systems, Business Analytics, Data Science, Digital Transformation, and Strategic Information Systems.
2. Contribute to real-world projects and internships involving ERP, SCM, CRM, e-commerce, and global information systems.
3. Apply knowledge in interdisciplinary fields, bridging engineering, business, and IT for innovation and problem-solving.
4. Prepare for professional roles in consulting, business analysis, system design, and digital business management within global organizations.

LITERATURE:

Reading Textbook 1

1. Kenneth C. Laudon; Jane Price Laudon, (2020), "**Management information systems: managing the digital firm**" Pearson Education, ISBN: 9780135191798,0135191793.

Reference Textbook 2

2. James A. O'Brien; George M. Marakas, "**Management Information Systems**", McGraw-Hill Irwin, ISBN 978-0-07-337681-3

Course assessments Criteria

<table>
<tr><th>Assessment</th><th>Components</th><th>Details</th><th>Marks</th><th>Assessment type</th><th>Marks</th><th>Total</th></tr>
<tr><td rowspan="5">Continuous Assessment1 & Midterm Examination</td><td>Assignment1 (Handwritten)</td><td>Theory and Application-based Questions</td><td>10 Marks</td><td>Individual</td><td rowspan="2">15 Marks</td><td rowspan="5">30 Marks</td></tr>
<tr><td>Participation in class practice sessions (Week 2 to Week 6)</td><td>Taking an active part in practice sessions</td><td>5 Marks (1 mark for each class)</td><td>Individual</td></tr>
<tr><td>Research Report</td><td>Research Report submission</td><td>5 Marks</td><td>Group</td><td rowspan="3">15 Marks</td></tr>
<tr><td rowspan="2">Project / Case Analysis / Class Presentation / Research</td><td>PPT preparation</td><td>5 Marks</td><td>Group</td></tr>
<tr><td>Oral presentation</td><td>5 Marks</td><td>Individual (presented in a group)</td></tr>
<tr><td colspan="7">Note: Follow all the instructions given by the Professor and adhere to timelines</td></tr>
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<tr><th>Assessment</th><th>Components</th><th>Details</th><th>Marks</th><th>Assessment type</th><th>Marks</th><th>Total</th></tr>
<tr><td rowspan="5">Continuous Assessment2 & End-term Examination</td><td>Assignment2 (Handwritten)</td><td>Theory and Application-based Questions</td><td>10 Marks</td><td>Individual</td><td rowspan="2">15 Marks</td><td rowspan="5">30 Marks</td></tr>
<tr><td>Participation in class practice sessions (Week 9 to Week 13)</td><td>Taking an active part in practice sessions</td><td>5 Marks (1 mark for each class)</td><td>Individual</td></tr>
<tr><td>Research Report</td><td>Research Report submission</td><td>5 Marks</td><td>Group</td><td rowspan="3">15 Marks</td></tr>
<tr><td rowspan="2">Project / Case Analysis / Class Presentation / Research</td><td>PPT preparation</td><td>5 Marks</td><td>Group</td></tr>
<tr><td>Oral presentation</td><td>5 Marks</td><td>Individual (presented in group)</td></tr>
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</table>

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<tr><td rowspan="2">Final Examination</td><td>Multiple Choice Questions</td><td>20 MCQs * 1 mark each</td><td>20 Marks</td><td>Individual</td><td rowspan="2">40 Marks</td><td rowspan="2">40 Marks</td></tr>
<tr><td>Project / Case Analysis / Research Presentation /</td><td>PPT preparation + Oral presentation / PPT or Report submission</td><td>20 Marks (10 + 10)</td><td>Group (PPT)
Individual (Presentation)</td></tr>
<tr><td colspan="7">Note: Follow all the instructions given by the Professor and adhere to timelines</td></tr>
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COURSE CALENDAR

Session (Day, Date) Batch day and time in the week (As mentioned under **Lessons** on page 1)	Title	Description	Text Book	Expected Learning Hours		
				LECTURE	*PRACTICE*	*SIS*
1	Information Systems in Global Business Today	• The Role of Information Systems in Business Today • Perspectives on Information Systems	Reading Textbook 1; Chapter 1 – 1.1 & 1.2; Page 32 - 66	*2*	*1*	*6*
2	Achieving Operational Excellence and Customer Intimacy: Enterprise Applications	• Enterprise Systems: Enterprise Resource Planning	Reading Textbook 1; Chapter 9 – 9.1 & 9.2; Page 366 – 371 + Notes	*2*	*1*	*6*
3	Achieving Operational Excellence and Customer Intimacy: Enterprise Applications	• Supply Chain Management Systems	Reading Textbook 1; Chapter 9 – 9.1 & 9.2; Page 372 - 379	*2*	*1*	*6*
4	Achieving Operational Excellence and Customer Intimacy: Enterprise Applications	• Customer Relationship Management Systems	Reading Textbook 1; Chapter 9 – 9.3 & 9.4; Page 381 - 386	*2*	*1*	*6*
5	Achieving Operational Excellence and Customer Intimacy: Enterprise Applications	• Enterprise Applications: New Opportunities and Challenges	Reading Textbook 1; Chapter 9 – 9.3 & 9.4; Page 387 - 396	*2*	*1*	*6*
6	E-commerce: Digital Markets, Digital Goods	• E-commerce and the Internet, Business and Technology • The Mobile Digital Platform and Mobile E-commerce • Building an E-commerce Presence	Reading Textbook 1; Chapter 10 – 10.1 & 10.2; Page 400 – 430; Chapter 10 – 10.3 & 10.4; Page 431 - 445	*2*	*1*	*6*

7	Project / Case Analysis / Class Presentation / Research	Project / Case Analysis / Class Presentation / Research	Reading Textbook 1; Reference Textbook 2	2	1	6
8	Midterm Examination	Midterm Examination	Reading Textbook 1; Reference Textbook 2	2	1	6
9	Information Systems, Organizations, and Strategy	• Using Information Systems to Achieve Competitive Advantage • Using Systems for Competitive Advantage: Management Issues	Reading Textbook 1; Chapter 3 – 3.3 & 3.4; Page 123 - 147	2	1	6
10	Global E-business and Collaboration	• Business Processes and Information Systems • Types of Information Systems	Reading Textbook 1; Chapter 2 – 2.1 & 2.2; Page 70 - 107	2	1	6
11	Enhancing Decision Making	• Decision Making and Information Systems • Business Intelligence in the Enterprise	Reading Textbook 1; Chapter 12 – 12.1 & 12.2; Page 482 - 500	2	1	6
12	Building Information Systems	• Systems as Planned Organizational Change • Overview of Systems Development	Reading Textbook 1; Chapter 13 – 13.1 & 13.2	2	1	6
13	AI Technologies and the use of AI Tools for businesses	• Introduction to AI in Business • Core AI Technologies & Tools for Business	Classroom teaching + Learning materials	2	1	6
14	Project / Case Analysis / Class Presentation / Research	Project / Case Analysis / Class Presentation / Research	Reading Textbook 1; Reference Textbook 2	2	1	6
15	End-term Examination	End-term Examination	Reading Textbook 1; Reference Textbook 2	2	1	6
16-17	Final Examination	Final Examination	Reading Textbook 1; Reference Textbook 2	2	1	6

Course assessment schedule

#		Session Week															Points
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
1	Continuous Assessment (1 & 2) Midterm & End-term						10							10			10+ 10
2	Participation in the practice sessions and activities, Pre-Midterm & Pre-End-term		1	1	1	1	1			1	1	1	1	1			5+5
3	Midterm Exam							5	10								15
4	EndTerm Exam														5	10	15
	Final Exam																40
	Total																***100***

Class sessions - Class sessions will comprise a blend of lectures, conceptual inputs, interactive discussions, practical application of theories, skill development exercises, case analyses, situational problem-solving, quizzes, and collaborative group activities

Practice – Practice sessions are designed to reinforce students' understanding by facilitating the practical application of concepts introduced during lectures.

In-class assessments are designed to prepare students for midterm and final evaluations, while measuring the level of competence they have attained in the subject matter. Their purpose is to diagnose potential gaps in understanding and performance, thereby enabling timely and specific adjustments to the teaching–learning process.

Home assignments – Home assignments are intended to consolidate concepts and materials covered during in-class activities, while encouraging students to expand their knowledge through engagement with diverse resources. These assignments also provide opportunities to practice specific skills and foster both independent and collaborative learning, thereby enhancing students' capacity to explore and critically examine related issues.

SIS (Student Independent Study) – Student Independent Study (SIS) encompasses both group-based and individual learning activities that students undertake autonomously. These activities require the application of knowledge and skills acquired during class sessions to complete assigned tasks and assessments. Teachers will provide guidance and support as needed, primarily during designated office hours.

TSIS (Teacher Supervised Student Independent Study) – student self-made project/activities/assignments/presentations.

Mid-term test/exam – A diagnostic assessment designed to evaluate students' progress, strengths, and areas for improvement, to reinforce learning, and prepare them for the final examination. This assessment includes assignments, research activities, and presentations.

End-term test/exam – A diagnostic assessment designed to evaluate students' progress, strengths, and areas for improvement, to reinforce learning, and prepare them for the final examination. This assessment includes assignments, research activities, and presentations.

Final examination

A summative assessment is designed to evaluate the extent to which students have successfully achieved the intended learning objectives of the course.

Grading policy:

Intermediate attestations (on the 7th/8th and 14th/15th week) join topics of all lectures, practice, SIS, TSIS, and materials for reading discussed at the time of attestation. The maximum number of points available for attendance, activity, SIS, TSIS, and laboratories in each attestation is 30 points.

The final exam joins and generalizes all course materials and is conducted in a complex form with research, presentations, and MCQs. The maximum number of points is 40. At the end of the semester, students receive an overall total grade (summarized index of their work during the semester) according to the conventional KBTU grade scale.

Attention!

1. If the student misses more than 30% of lessons without a plausible reason, then the student receives an «F (Fail)» grade.
2. If, for two attestations, a student receives 29.4 or less points, this student is not accepted to the final exam, and for the entire course, they get an «F (Fail)» grade;
3. If a student receives 9.4 or less points on the final exam, then independently of how many points he (she) received for the two attestations, in whole, he (she) gets an «F (Fail)» grade;

In the case of missing or being late for the final exam without a plausible reason, independently of how many points he (she) received for the two attestations, in whole, he (she) gets an «F (Fail)» grade.

Academic Policy:

1. Cheating, duplication, falsification of data, and plagiarism are prohibited under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute "participation." Participation involves reading the assigned materials, coming to class prepared to ask questions, and actively engaging in discussions.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
4. Students must come to class on time, and no food or drinks are allowed during classes.
5. Students are responsible for making up any work they missed.
6. Make-up tests in case of absence will not usually be allowed.
7. Mobile phones and Laptops must always be switched off in class unless you are asked to participate in an academic activity. Unauthorized use of mobile/laptop devices or sleeping during class will result in the loss of attendance/assessments/grades.
8. Students should always show tolerance, consideration, and mutual support toward others.

Students are encouraged to

1. Consult the teacher on any issues related to the course.
2. Make up work within a week for a valid reason without any grade deductions.
3. Propose improvements to the academic process.
4. Track down their continuous rating throughout the semester.

Academic Integrity

Whenever you put your name on a piece of work for the source, you assert that it is your work. If this is not the case, it is plagiarism. In proper citation format, you must report the source of information used in a paper. Quotation marks and page references must denote direct quotes. Plagiarism is considered a form of cheating and will be addressed in accordance with KBTU's policies and procedures. Cheating on exams and assignments (or failing to provide proper citations) will not be tolerated and will be addressed in accordance with KBTU's policies and procedures.

Students are required:

1. To be respectful to the teacher and other students;
2. To meet the assignments and academic deadlines;
3. To be prepared and actively participate in classroom work;
4. To attend all classes and tests/exams as per the schedule

Associate Professor of SITE *Dr. Avinash BM*

Minutes # 1 of the School of Information Technology and Engineering meeting on August 18, 2025